

Ryan Tsai

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EDUCATION

BE (Hons) Software Engineering, University of Auckland; 2024 – 2027

- *Grade:* 8.18/9.00 GPA
- *Awards:* Dean's List 2025 (Top 5%)
- *Relevant Coursework:* Data Structures & Algorithms, Software Quality Assurance, Database Systems
- *Clubs:* SESA, WDCC, DEVS

EXPERIENCE

Thinking2Morrow, English and Mathematics Tutor

Feb 2024 - Present

- Tutor students from Year 1 to 13 across NCEA, IGCSE, Cambridge and IB curricula.
- Delivered personalized lessons and feedback, both in individual and group settings (up to 12 students).
- Developed key professional skills: time management, communication, collaborative planning.

SKILLS

Languages: TypeScript, JavaScript, Java, Python, C, VHDL

Frameworks & Libraries: Node.js, JavaFX, Express.js, React

Tools and Skills: Git, GitHub Flow, Maven, MongoDB, SQLite, Firebase, Cloudflare R2, PITest, JUnit

PROJECTS

Astral Images

- Created a full-stack cloud image hosting website with **Cloudflare R2**, **React** and **Express**
- Integrated **Firebase** authentication and user management.
- Implement CI/CD pipelines for deployment using Vercel's hosting service.

Ferrous Fellow

- Designed a toy robot that recommends music for children based off visual and verbal cues.
- Integrated **TensorFlow Lite** and **OpenCV** onto a **Raspberry Pi Pico**
- Integrated **OpenAI API** to recommend music based on collected speech and facial data

VALORANT API Wrapper

- Created a **TypeScript** wrapper using **Express.js**, enabling headless access to game data without requiring an active client session.
- Created an abstraction layer to map and fetch client-locked data, allowing external access of user data without an active game instance.

MeowPad

- Designed a custom mechanical number pad from concept to manufactured product, using **KiCad** for schematic and PCB design and **AutoCAD Inventor** for case design.
- Developed custom **C firmware** using the QMK library to support **Arduino Pro Micro** footprint MCUs, integrating an OLED display and EC11 rotary encoder.